

Heavy Metal Concentration Report

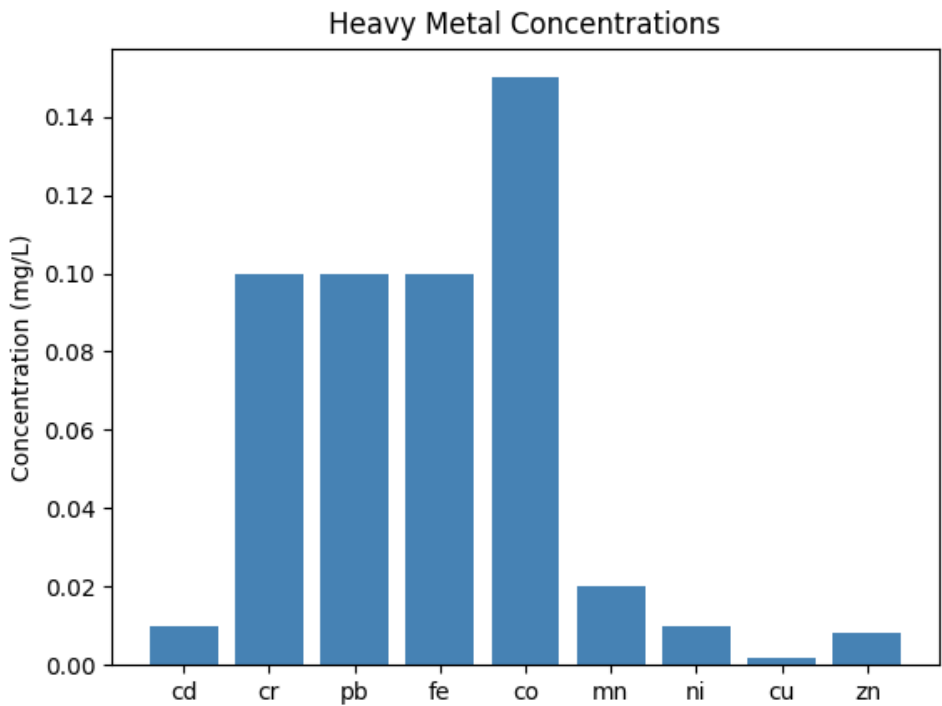
Sample Collection Summary

Latitude: 22.41
Longitude: 88.85
Source: Lake

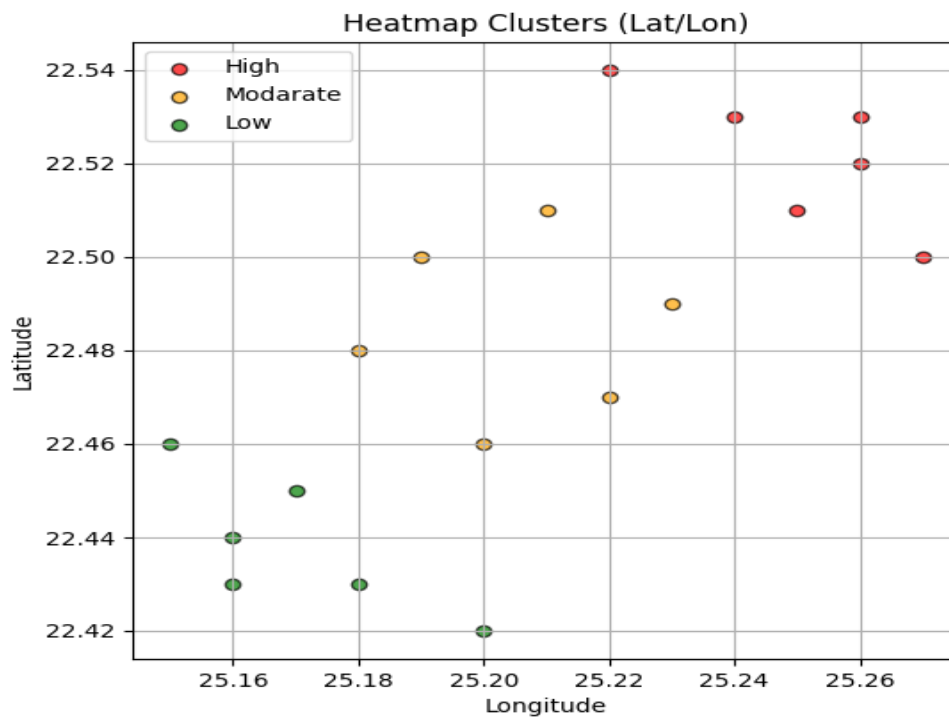
Indices Summary

Contamination Degree (Cd): 2.409
Health Exposure Index (HEI): 11.409
Heavy Metal Pollution Index (HMPI): 268.349
*All the formulae and definations have been described in the generated report card.

Heavy Metal Concentrations



Heatmap Analysis



Anomaly Detection

Decision: WARN

- cr (0.1 mg/L) exceeds regulatory/warning threshold (0.05 mg/L)
- co (0.15 mg/L) exceeds regulatory/warning threshold (0.05 mg/L)

Health Impacts

- The high cadmium (Cd) levels in water can cause kidney damage, osteomalacia, osteoporosis, and is a probable human carcinogen.
- The very high lead (Pb) levels are extremely concerning and can lead to severe neurological damage, particularly in children affecting cognitive development and IQ, kidney damage, and cardiovascular problems.
- The elevated cobalt (Co) levels may contribute to cardiomyopathy (heart muscle disease), neurological issues, and thyroid dysfunction with chronic exposure.
- The elevated manganese (Mn) levels can lead to neurological effects (manganism) and developmental neurotoxicity, especially in infants and children, with chronic exposure.

Precautions

- Implement immediate and comprehensive water quality testing and monitoring programs to identify sources and accurately assess the full scope of heavy metal contamination.
- Invest in and deploy advanced water treatment technologies, such as reverse osmosis or ion exchange, capable of effectively removing cadmium, lead, cobalt, and manganese from the water supply.

- Issue urgent public health advisories recommending the avoidance of untreated tap water and providing immediate access to safe alternative drinking water sources for all affected communities.
- Initiate a comprehensive infrastructure audit and upgrade program to identify and replace lead service lines and other potential sources of metal leaching into the water system.
- Conduct thorough health risk assessments and long-term epidemiological studies to quantify the potential health impacts on exposed populations and inform targeted medical interventions.
- Review and strengthen existing national water quality standards and enforcement mechanisms to ensure strict adherence to safe limits for heavy metals.